Bounded Modular Recovery: Finite Certificates, Domain Control, and a Source Dependency Audit

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M580

October 7, 2026
Literature checked October 8, 2026 UTC

Abstract

We examine the pinned manuscript *Bounded recovery for modular spectral averages*, distinguishing its bounded-recovery mechanism from its global bicentralizer application. All eight TeX files are included in the source scope. We specify the standard-form domains, predual convention, weak normal-functional integrals, two norm ergodic limits, ordered fourth-moment expansion, polar cutoff, and joint spectral assembly. The local contribution is a proved finite-phase recovery certificate with separate spectral leakage, fourth-moment, polar-truncation, and kernel-approximation budgets. Three roots of unity suffice for exact second and fourth moments, and a four-wise-independent field design gives fewer than $81N^4$ candidates for $N > 1$ orbit samples. An explicit integrable Fourier kernel gives arbitrary fixed band enlargement greater than one, with a stated operator cost and a finite-time graph-norm error bound. These results make the local recovery step testable without supplying an ergodic convergence rate or independently verifying the source's global theorem. RamCS and domain-aware Ram representations organize the certificates; the spectral theorem, modular theory, randomization, and truncation remain classical ingredients. Contribution assessment: C3/5, finite-phase recovery with leakage and graph-norm error budgets gives a notable local extension while global rigidity and bicentralizer dependencies remain outside independent verification.

Contents

1 Pinned problem and exact mathematical contract **2**
1.1 Source identity and scope . . . 2
1.2 Spaces, domains, and conventions . . . 3
1.3 Quantifiers in recovery and rigidity . . . 3

2 Domain-correct Fourier and predual estimates **4**

3 Audit of averaging, fourth moments, and recovery **5**
3.1 Two distinct ordinary norm limits . . . 5
3.2 Initial representatives and the varying test . . . 6
3.3 Discretization and ordered fourth moments . . . 6
3.4 Polar truncation, relocalization, and selection . . . 7

4 Finite-phase recovery with an explicit error budget **7**
4.1 What is made finite, and what is not 8
4.2 Polynomial-size exact phase families 9

5 Explicit filters, bandwidth, and finite-time control **9**
5.1 A compact trapezoidal multiplier 9
5.2 Arbitrarily small fixed enlargement 10
5.3 Finite time preserves a graph-norm certificate 11
5.4 What finite quadrature would additionally require 11

6 Spectral rigidity and bicentralizer dependency assembly **12**
6.1 From shrinking bands to an interval 12
6.2 The symmetric pairing and its sign 12
6.3 Spectral geometry and exclusion of modular eigenvalues 13
6.4 Where the global application imports theorems 13

7 Specific Ram correspondences and their limits **14**
7.1 RamCS: the exact normalization bridge 14
7.2 Separate approximation budgets 14
7.3 Domain transport and false finite conclusions 15
7.4 What is classical and what is contributed 15

8 Current literature, comparisons, and open dependencies **15**
8.1 Why scalar centralizer cannot simply be dropped 16
8.2 Nonzero versus nonscalar witnesses 17

9 Verification and contribution boundary **17**

1 Pinned problem and exact mathematical contract

1.1 Source identity and scope

The object studied is OpenAI's September 23, 2026 manuscript *Bounded recovery for modular spectral averages*, catalogue ID M580, at catalogue commit [1]

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The catalogue's applicability level is not a contribution rating. The local source root is

`C:/Users/mural/OneDrive/Documents/OpenAI-722-math/`
`_sources/openai-math/preprints/`
`Bounded-recovery-for-modular-spectral-averages-September-23-2026/build/`

The complete source scope consists of these eight files:

- `paper.tex`, `preamble.tex`, and `figures/bounded-recovery.tex`;
- `sections/01-introduction.tex` and `sections/02-bicentralizers.tex`;
- `sections/03-modular-averaging.tex`;
- `sections/04-bounded-recovery.tex`;
- `sections/05-spectral-rigidity.tex`.

The bibliography `references.bib` was also read. There is no separate mathematical appendix or unexamined substantive input in these eight files. The local snapshot has no Git repository metadata; the commit attribution is the catalogue pin, supplemented by per-file SHA-256 records, not an independent Git checkout verification.

1.2 Spaces, domains, and conventions

Let M be a von Neumann algebra with faithful normal state ϕ, $\phi(1) = 1$. Work in its standard Hilbert space $H = L^2(M)$, with cyclic separating unit vector $\xi = \phi^{1/2}$. Inner products are linear in the *first* variable. Left multiplication is $L(a)h = ah$; right multiplication is $R(a)h = ha = JL(a)^*Jh$. Thus R is an antirepresentation: $R(a)R(b) = R(ba)$. Set

$$S = J\Delta_\phi^{1/2}, \qquad D = \log \Delta_\phi, \qquad U_t = e^{itD}, \qquad \sigma_t = \sigma_t^\phi.$$

Here S is the closed Tomita operator, the closure of $a\xi \mapsto a^*\xi$, and J is conjugate linear with $J\xi = \xi$. The spectral theorem defines the self-adjoint D, with spectral resolution E_D, and

$$\begin{aligned} \mathrm{Dom}(D) &= \{h : \int_{\mathbb{R}} u^2 \, d\,\|E_D(u)h\|^2 < \infty\}, \\ \mathrm{Dom}(e^{D/2}) &= \{h : \int_{\mathbb{R}} e^u \, d\,\|E_D(u)h\|^2 < \infty\}. \end{aligned} \tag{1.1}$$

In these integrals $d\,\|E_D(u)h\|^2$ denotes the positive scalar spectral measure of h, not a derivative of an arbitrarily chosen coordinate. For every $a \in M$, $a\xi \in \mathrm{Dom}(e^{D/2})$ and

$$U_t a\xi = \sigma_t(a)\xi, \quad e^{D/2}a\xi = Ja^*\xi = \xi a, \quad \|a\xi\|^2 = \phi(a^*a), \quad \left\|e^{D/2}a\xi\right\|^2 = \phi(aa^*). \tag{1.2}$$

In particular, arbitrary Hilbert vectors need not be in this domain, even though algebra vectors are. Also $JDJ = -D$ and $Je^{D/2}J = e^{-D/2}$ on the transported domains. Conjugate linearity gives $JU_tJ = U_t$, not U_{-t}.

There is no spatial boundary-value problem here. The admissibility conditions are operator domains, spectral support, and the state normalization. Compact bands are closed real intervals; kernel integrals are over real time with Lebesgue measure. Neither a complex contour shift nor integration by parts at an undeclared boundary is used.

We use the source's predual actions

$$(w\phi)(y) = \phi(yw), \qquad (\phi w)(y) = \phi(wy), \qquad \|\rho\|_{M_*} = \sup_{\|y\|\le 1} |\rho(y)| \,. \tag{1.3}$$

The modular centralizer is $M_\phi = \{a : \sigma_t(a) = a \ \forall t\}$. The source assumes $M_\phi = \mathbb{C}1$ for recovery. This is not the condition $(M_\phi)' \cap M = \mathbb{C}1$ called an irreducible centralizer.

1.3 Quantifiers in recovery and rigidity

Fix a free ultrafilter ω on $\mathbb{N}$ once, and let

$$m_t f(t) = \lim_{k\to\omega} \frac{1}{2k} \int_{-k}^{k} f(t)\, dt, \quad f \in C_b(\mathbb{R}).$$

This mean is positive, normalized, translation invariant and reflection invariant. It is not a probability density on $\mathbb{R}$. For bounded continuous H-valued F, put $\|F\|_{\mathrm{av}} = (m_t \|F(t)\|^2)^{1/2}$.

The source recovery statement is: for each *fixed* $T \in \mathcal{B}(H)$ and $s \in \mathbb{R}$, if $\delta_n > 0$ tends to zero, $\|h_n\| = 1$, $h_n \in E_D([s-\delta_n, s+\delta_n])H$, and

$$\limsup_n m_t \|TU_t h_n\|^2 > 0, \tag{1.4}$$

then a subsequence n_j and $v_j \in M$ exist with constants $C_* < \infty$, $\eta > 0$ such that, for *every* j,

$$\|v_j\| \le C_*, \quad \|Tv_j\xi\| \ge \eta, \quad v_j\xi \in E_D([s - 4\delta_{n_j}, s + 4\delta_{n_j}])H. \tag{1.5}$$

It does not specify a uniform averaging time, an algorithmic complexity, or constants uniform in s, T, and the positive lower bound. It does not assert that $v_j\xi$ approximates the original h_{n_j}.

The source rigidity premise is stronger and separate. A u-continuous ϕ-preserving action $b : \mathbb{R} \curvearrowright M$ must satisfy, for every $s \in \mathbb{R}$, every uniformly bounded sequence $w_n \in M$, and every $a \in M$,

$$\|\phi w_n - e^s w_n \phi\|_{M_*} \to 0 \quad \Longrightarrow \quad w_n a - b_s(a) w_n \to 0 \quad \text{strongly.} \tag{1.6}$$

Under scalar centralizer the source concludes $b_s = \mathrm{id}$ for every s. Its final application is $\mathrm{BC}(M, \phi) = \mathbb{C}1$ for a type III_1 factor with separable predual and *any* faithful normal state. The original state's centralizer need not be scalar: scalarity is obtained after a self-bicentralizer reduction. The results below do not independently certify that global conclusion.

Explicitly, $a \in \mathrm{BC}(M, \phi)$ means that for every sequence $w_n \in M$ with $\sup_n \|w_n\| < \infty$ and $\|w_n\phi - \phi w_n\|_{M_*} \to 0$, the commutators $[a, w_n]$ tend to zero strongly. Here strongly means convergence on each vector in H, not operator-norm convergence. The source uses the standard equivalence with the unitary formulation: for every $\varepsilon > 0$ there is $\delta > 0$ such that every unitary u with $\|u\phi - \phi u\|_{M_*} < \delta$ satisfies $\|(u^*au - a)\xi\| < \varepsilon$. This equivalence is an established bicentralizer input, not a new consequence of the local filters.

2 Domain-correct Fourier and predual estimates

Proposition 2.1 (Weak Fourier filter and graph control)**.** *For $g \in L^1(\mathbb{R})$ define $f(u) = \int_{\mathbb{R}} g(t)e^{itu}\,dt$. For $y \in M$ there is a unique $\mathcal{F}_g(y) \in M$ satisfying*

$$\rho(\mathcal{F}_g(y)) = \int_{\mathbb{R}} g(t)\rho(\sigma_t(y))\,dt \quad (\rho \in M_*). \tag{2.1}$$

It satisfies

$$\|\mathcal{F}_g(y)\| \le \|g\|_1 \|y\|, \qquad \mathcal{F}_g(y)\xi = f(D)y\xi, \qquad e^{D/2}\mathcal{F}_g(y)\xi = f(D)e^{D/2}y\xi. \tag{2.2}$$

If $g, \widetilde{g} \in L^1$ and $\tau = \|g - \widetilde{g}\|_1$, then

$$\begin{aligned} \left\|(\mathcal{F}_g - \mathcal{F}_{\widetilde{g}})(y)\xi\right\| &\le \tau \|y\xi\|, \\ \left\|e^{D/2}(\mathcal{F}_g - \mathcal{F}_{\widetilde{g}})(y)\xi\right\| &\le \tau \left\|e^{D/2}y\xi\right\|. \end{aligned} \tag{2.3}$$

Proof. The right side of (2.1) is a bounded linear functional on M_* with norm at most $\|g\|_1 \|y\|$, hence is represented in $(M_*)^* = M$. Vector functionals and the Bochner integral in H identify its value on ξ with $\int g(t)U_t y\xi\,dt = f(D)y\xi$. No norm continuity of $t \mapsto \sigma_t(y)$ is necessary. A bounded Borel multiplier $f(D)$ preserves $\mathrm{Dom}(e^{D/2})$: the weighted spectral integral is bounded by $\|f\|_\infty^2$ times that for $y\xi$. Multiplication of spectral functions proves the last equality in (2.2). Apply the same reasoning to $g - \widetilde{g}$ and use $\left\|f - \widetilde{f}\right\|_\infty \le \tau$ to prove (2.3). □

For $f \in C_c^\infty$ the required kernel is $g(t) = (2\pi)^{-1} \int f(u)e^{-itu}\,du$. The proposition permits a larger class of filters than the source's smooth sufficient class, but does not permit a discontinuous spectral indicator without proving an integrable kernel representation.

Proposition 2.2 (Localized predual certificate)**.** *For $w \in M$, $s \in \mathbb{R}$, and $c = e^{s/2}$,*

$$\|\phi w - e^s w\phi\|_{M_*} \le (1+c)\left\|(e^{D/2} - c)w\xi\right\|. \tag{2.4}$$

If $w\xi \in E_D([s-d, s+d])H$, then

$$\|\phi w - e^s w\phi\|_{M_*} \le (1+c)c(e^{d/2} - 1)\|w\xi\|. \tag{2.5}$$

Proof. Set $\eta = \xi w - cw\xi$. By (1.2), this is the vector on the right of (2.4); $J\eta = w^*\xi - c\xi w^*$. For $\|y\| \le 1$, begin with $\phi(wy) = \langle y\xi, w^*\xi\rangle$. Replacing $w^*\xi$ by $c\xi w^*$ costs at most $\|\eta\|$. The identity $\langle y\xi, \xi w^*\rangle = \langle y(\xi w), \xi\rangle$ follows from commuting left and right actions. Replacing ξw by $cw\xi$ now costs at most $c\|\eta\|$ and leaves $c^2\phi(yw)$. Taking the supremum proves (2.4). On the indicated band, $\sup\left|e^{u/2} - c\right| = c(e^{d/2} - 1)$, proving the second assertion. □

Small unweighted vector error is insufficient to use this estimate. For example, if $Dh_N = Nh_N$, $\|h_N\| = 1$, then $e^{-N/4}h_N \to 0$ but $\left\|e^{D/2}e^{-N/4}h_N\right\| = e^{N/4}$. This elementary spectral model illustrates the domain obstruction; it is not asserted to model a scalar-centralizer factor.

If $a\xi$ has compact modular spectral support K, choose a smooth compact multiplier f equal to $e^{-u/2}$ near K and set $a^\flat = \mathcal{F}_g(a)$. Then $\xi a^\flat = e^{D/2}f(D)a\xi = a\xi$. This bounded right multiplier is the source's legitimate replacement for the unjustified formal operator $R(e^{-D/2}a)$.

3 Audit of averaging, fourth moments, and recovery

This section reconstructs the logical dependencies of source labels `mod:means`, `mod:average`, and `rec:bounded`. Reconstruction of these steps is not a declaration that all imported operator-algebra theorems or the final global application have been independently verified.

3.1 Two distinct ordinary norm limits

Assume $M_\phi = \mathbb{C}1$ and write $A_k(y) = (2k)^{-1}\int_{-k}^{k}\sigma_t(y)\,dt$. The unitary mean ergodic theorem gives an H-norm limit of $A_k(y)\xi$. The operator averages have norm at most $\|y\|$. For fixed u their translation defect is at most $|u|\,\|y\|/k$ for large k. Every weak operator cluster point is therefore fixed by σ and is scalar, with scalar $\phi(y)$. Hence

$$A_k(y)\xi \longrightarrow \phi(y)\xi \quad \text{in } H. \tag{3.1}$$

This is a statement for each fixed y; it is not operator-norm convergence.

For $\psi \in M_*$, the action $t \mapsto \psi\circ\sigma_{-t}$ is norm continuous. Indeed, a positive normal functional is a standard-form vector functional; strong continuity of U_t and $\|\omega_v - \omega_w\| \le (\|v\| + \|w\|)\,\|v - w\|$ prove the claim, and linear decomposition handles arbitrary functionals. Its averages are Bochner integrals in M_*. Let

$$N = \overline{\operatorname{span}}^{\|\cdot\|}\{\rho\circ\sigma_u - \rho : \rho \in M_*,\ u \in \mathbb{R}\}.$$

The annihilator $N^\perp$ is $M_\phi = \mathbb{C}1$. Hahn–Banach therefore gives $N = \{\rho : \rho(1) = 0\}$. Interval averages tend to zero on each coboundary by the same boundary estimate, and then on its closed span by contractivity. Since they fix ϕ, it follows that

$$\psi_k = \frac{1}{2k}\int_{-k}^{k}\psi\circ\sigma_{-t}\,dt \longrightarrow \psi(1)\phi \quad \text{in } M_*. \tag{3.2}$$

Using weak convergence instead here would not control a varying test operator. Scalarity is what identifies the entire annihilator, so this argument must not be transferred to a general centralizer unchanged.

For $S \in \mathcal{B}(H)$ define $P(S)$ by the bounded sesquilinear form $\langle P(S)h, k\rangle = m_t\langle U_t^* S U_t h, k\rangle$. Positivity, unitality, and contraction follow from the mean. Translation invariance makes $P(S)$ commute with E_D, while weak closure of $L(M)$ gives

$$P(L(a)) = \phi(a)1, \qquad m_t \left\|SU_t h\right\|^2 = \langle P(S^*S)h, h\rangle. \tag{3.3}$$

Normality of P is neither established nor needed. Countable additivity must be used for E_D, not silently attributed to m or P.

3.2 Initial representatives and the varying test

Starting from (1.4), a subsequence has averages at least $4\gamma > 0$. Density of $M\xi$ followed by a contraction filter fixing $[s - \delta_n, s + \delta_n]$ gives normalized $y_n\xi$ with support radius $d_n = 2\delta_n$, $\|y_n\xi - h_n\| \to 0$, and averages at least 2γ. The seminorm triangle inequality justifies this preservation of the defect. After discarding finitely many indices, $d_n \le 1$ and

$$\phi(y_n^* y_n) = 1, \qquad \phi(y_n y_n^*) \le e^{s+d_n} \le e^{s+1}. \tag{3.4}$$

There is no uniform bound on $\|y_n\|$, nor is one used.

Fix one index and write $y = y_n$. Put

$$A_k = A_k(y^*y), \quad B_k = A_k(yy^*), \quad \psi(x) = \phi(y^*xy), \quad \psi_k(x) = \frac{1}{2k}\int_{-k}^{k} \phi(\sigma_t(y)^* x \sigma_t(y))\, dt.$$

Then ψ is a normal state, $\|B_k\| \le \|y\|^2$, and $\phi(B_k) = \phi(yy^*)$. The two limits above separately imply

$$\phi(A_k^2) \to 1, \qquad |\psi_k(B_k) - \phi(B_k)| \le \|\psi_k - \phi\|_{M_*} \|y\|^2 \to 0. \tag{3.5}$$

The order of quantifiers is $\forall n\ \exists k_0(n)\ \forall k \ge k_0(n)$ for the moment estimates, not $\exists k_0\ \forall n$. The ultrafilter-large set where the finite-time defect exceeds $3\gamma/2$ meets this cofinite set. Thus one time can be chosen for each n with both properties. This use of an ultrafilter is qualitative; it supplies no stopping criterion for searching through times.

3.3 Discretization and ordered fourth moments

The modular orbit is bounded and strong-star continuous. Consequently the three scalar integrands

$$\|T\sigma_t(y)\xi\|^2, \quad \phi(z_t^* z_t z_u^* z_u), \quad \phi(z_t^* z_u z_u^* z_t), \qquad z_t = \sigma_t(y),$$

are continuous on the selected compact interval or its square. A common Riemann partition, with probability weights p_i, approximates all three; the double integrals require the product weights $p_i p_j$. For $z_i = \sigma_{t_i}(y)$ let

$$A = \sum_i p_i z_i^* z_i, \quad B = \sum_i p_i z_i z_i^*, \quad a_T = \sum_i p_i \|Tz_i\xi\|^2. \tag{3.6}$$

The source arranges $a_T \ge \gamma$ and $\phi(A^2) + \sum_i p_i \phi(z_i^* B z_i) \le C = 4 + e^{s+1}$. It does not require a uniform partition size.

For independent uniform circle phases ϵ_i and $z = \sum_i \sqrt{p_i}\epsilon_i z_i$, direct expansion gives

$$\mathbb{E}\phi((z^*z)^2) = \phi(A^2) + \sum_i p_i \phi(z_i^* B z_i) - \sum_i p_i^2 \phi((z_i^* z_i)^2). \tag{3.7}$$

The pairings are $(i = j, k = l)$ and $(i = l, j = k)$ in $z_i^* z_j z_k^* z_l$; their all-equal overlap must be subtracted once. No trace property is used. Replacing the second term by $\phi(B^2)$ would be unjustified for a general state. Each subtracted term is nonnegative, so the source obtains $\mathbb{E}\phi(|z|^4) \le C$.

3.4 Polar truncation, relocalization, and selection

For $z = u|z|$ and $K > 0$, put $z^{[K]} = u \min(|z|, K)$. Then

$$\left\|z^{[K]}\right\| \le K, \qquad \left\|(z - z^{[K]})\xi\right\|^2 = \phi((|z| - K)_+^2).$$

The source bounds its expectation by C/K^2. This is a one-sided state-L^2 estimate. An operator-norm estimate or a small adjoint error does not follow from it. Spectral localization is generally destroyed by this nonlinear cutoff, so the final filter is essential.

Write $f_\rho(u) = f_0((u - s)/\rho)$ where $f_0 = 1$ on $[-1, 1]$, $0 \le f_0 \le 1$, and $\operatorname{supp} f_0 \subset [-2, 2]$. The inverse kernel is $g_\rho(t) = \rho e^{-ist} g_0(\rho t)$, hence

$$\|g_\rho\|_1 = \|g_0\|_1 =: G_0. \tag{3.8}$$

Set $v = \mathcal{F}_{g_{d_n}}(z^{[K]})$. The filter fixes $z\xi$ and is a contraction on H, so

$$(\mathbb{E}\,\|Tv\xi\|^2)^{1/2} \ge \sqrt{\gamma} - \|T\|\sqrt{C}/K.$$

One $K \ge 2\|T\|\sqrt{C/\gamma}$ works for every n. The operator norm is at most $G_0 K$, and the output radius is $2d_n = 4\delta_n$. The source selects a maximizing phase point: the polar cutoff is norm continuous because it equals $zq_K(z^*z)$ for continuous $q_K(r) = \min(1, K/\sqrt{r})$, with $q_K(0) = 1$. Compactness of the finite phase torus justifies the maximum. Section 4 replaces this last compact search by an exact finite alphabet and quantifies errors.

4 Finite-phase recovery with an explicit error budget

This local theorem does *not* assume a scalar centralizer, an invariant mean, an ergodic theorem, a shrinking-band limit, or a bicentralizer law. Its hypotheses are finite data and certified bounds. Producing these data uniformly along the source's sequence still requires the previous section.

Lemma 4.1 (Sharp scalar cutoff majorant)**.** *For $p > 2$, $K > 0$, and $r \ge 0$,*

$$(r - K)_+^2 \le c_p K^{2-p} r^p, \qquad c_p = \frac{4(p-2)^{p-2}}{p^p}. \tag{4.1}$$

In particular $c_4 = 1/16$. The scalar constant is best possible. The $p = 4$ bound is classical: Haagerup–Musat [3, equations (2.18)–(2.20)] give $|G_K(t)| \le t^2/(4K)$ and the resulting operator cutoff bound with coefficient $1/(16K^2)$.

Proof. For $r \le K$ the assertion is immediate. For $x = r/K > 1$, maximize $(x-1)^2/x^p$. Its logarithmic derivative is $2/(x-1) - p/x$, with the unique zero $x = p/(p-2)$; the function vanishes at 1 and at infinity. Substitution gives c_p. Equality at this x proves optimality of the scalar majorant. Functional calculus and a positive state preserve the resulting inequality for $|z|$. □

This recovers a sharper classical inequality than the one used in the pinned source's displayed estimate; it is not a new cutoff constant.

Theorem 4.2 (Finite-phase recovery certificate)**.** *Let $z_1, \dots, z_N \in M$, $p_i \ge 0$, $\sum_i p_i = 1$, and $T \in \mathcal{B}(H)$. Define A, B, a_T by (3.6) and define*

$$C_4 = \phi(A^2) + \sum_i p_i \phi(z_i^* B z_i) - \sum_i p_i^2 \phi((z_i^* z_i)^2). \tag{4.2}$$

Suppose $g \in L^1$, $f(u) = \int g(t)e^{itu}dt$, $\|f\|_\infty \le 1$, *and put*

$$G = \|g\|_1, \qquad L_f^2 = \sum_i p_i \|(I - f(D))z_i\xi\|^2. \tag{4.3}$$

Let $\widetilde{g} \in L^1$, $\tau = \|\widetilde{g} - g\|_1$ *and* $K > 0$. *If*

$$b = \sqrt{a_T} - \|T\| \left(L_f + \frac{\sqrt{C_4}}{4K} + K\tau \right) > 0, \tag{4.4}$$

then some $\epsilon = (\epsilon_1, \dots, \epsilon_N) \in \{1, \zeta, \zeta^2\}^N$, *where* $\zeta = e^{2\pi i/3}$, *gives*

$$z_\epsilon = \sum_i \sqrt{p_i}\epsilon_i z_i, \quad v_\epsilon = \mathcal{F}_{\widetilde{g}}(z_\epsilon^{[K]}), \qquad \|v_\epsilon\| \le (G + \tau)K, \quad \|Tv_\epsilon\xi\| \ge b. \tag{4.5}$$

If $\widetilde{g} = g$ *and* $\operatorname{supp} f \subset I$, *then* $v_\epsilon\xi \in E_D(I)H$. *In general every realization satisfies* $\|E_D(\mathbb{R} \setminus \operatorname{supp} f)v_\epsilon\xi\| \le K\tau$.

Proof. Use independent uniform phases on the displayed three-point set. For an individual phase, $\mathbb{E}\epsilon^k = 0$ for $k = \pm 1, \pm 2$. In a second moment the exponents per phase are at most one in absolute value; in the fourth-moment expansion they are between -2 and 2. Thus precisely the same surviving pairings as for circle phases occur. Consequently

$$\mathbb{E}\|Tz_\epsilon\xi\|^2 = a_T, \qquad \mathbb{E}\phi(|z_\epsilon|^4) = C_4 \ge 0, \qquad \mathbb{E}\|(I - f(D))z_\epsilon\xi\|^2 = L_f^2.$$

By Lemma 4.1 and positivity, $\mathbb{E}\left\|(z_\epsilon - z_\epsilon^{[K]})\xi\right\|^2 \le C_4/(16K^2)$. For each phase, the exact identity

$$\mathcal{F}_g(z_\epsilon^{[K]})\xi - z_\epsilon\xi = f(D)(z_\epsilon^{[K]} - z_\epsilon)\xi + (f(D) - I)z_\epsilon\xi$$

and the triangle inequality in the finite Hilbert direct sum give an RMS error at most $\sqrt{C_4}/(4K) + L_f$. The kernel perturbation adds at most $K\tau$ for each realization, by Proposition 2.1. A second triangle inequality, now after applying T, proves $(\mathbb{E}\|Tv_\epsilon\xi\|^2)^{1/2} \ge b$. The maximum over 3^N realizations is at least this RMS, proving existence. The operator bound follows directly from the weak integral. The exact filtered vector has the asserted support, and its perturbation has norm at most $K\tau$, which proves the leakage conclusion. □

4.1 What is made finite, and what is not

The finite alphabet removes the need for an existence argument over a continuous torus and permits exhaustive deterministic selection once each filtered cutoff is evaluable. Its cost is 3^N evaluations, so no efficient algorithm or speedup is asserted. Rademacher signs are not an exact substitute: $\mathbb{E}\epsilon^2 = 1$ introduces additional pairings. Three-point phases match the required moments, not the entire distribution of the nonlinear cutoff.

It is safe to replace a_T by a certified lower bound, C_4 by a certified upper bound, and $L_f, \tau, \|T\|$ by upper bounds in (4.4). For a nonnegative interval upper bound on C_4 one may drop its negative overlap term, as the source does. Numerically subtracting nearly equal positive quantities without error control does not certify C_4.

If a bound $C_p \ge \mathbb{E}\phi(|z_\epsilon|^p)$ is available for any real $p > 2$, the identical proof replaces $\sqrt{C_4}/(4K)$ by $\sqrt{c_p C_p}\, K^{1-p/2}$. For $p \ne 4$ this moment bound is an *additional hypothesis*; the three-point phase rule does not identify it with the circle-phase pth moment. For p near two the cutoff error decreases very slowly, exposing the role of uniform integrability.

Corollary 4.3 (A conservative sampling guarantee)**.** *Under Theorem 4.2, put* $M_T = \|T\|\,(G+\tau)K$*. For* $0 < \eta < b$ *a uniform three-phase sample obeys*

$$\mathbb{P}\{\|Tv_\epsilon\xi\| > \eta\} \geq \frac{b^2-\eta^2}{M_T^2-\eta^2} =: q > 0. \tag{4.6}$$

Independent repetitions have failure probability at most $(1-q)^m$ *after* m *trials.*

Proof. For $X = \|Tv_\epsilon\xi\|^2$, $0 \leq X \leq M_T^2$ and $\mathbb{E}X \geq b^2$. If $p = \mathbb{P}(X > \eta^2)$, then $\mathbb{E}X \leq \eta^2(1-p) + M_T^2 p$. Rearrangement proves the estimate. The assumptions imply $b \leq M_T$, so the denominator is positive. Independence gives the last assertion. □

This is a mathematical probability bound, not an observed success rate. The filter cost and the moment estimate may make it very conservative.

4.2 Polynomial-size exact phase families

Proposition 4.4 (Four-wise-independent ternary design)**.** *For* $N > 1$*, let* r *be the least positive integer with* $q = 3^r \geq N$*, choose distinct* $x_1, \ldots, x_N \in \mathbb{F}_q$*, and run over* $(a_0, a_1, a_2, a_3) \in \mathbb{F}_q^4$*. The phases*

$$\epsilon_i(a) = \zeta^{\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_3}(a_0 + a_1x_i + a_2x_i^2 + a_3x_i^3)} \tag{4.7}$$

with uniformly distributed coefficient quadruples have the same second and fourth moments used in Theorem 4.2. That theorem and Corollary 4.3 therefore remain true on a multiset with $q^4 < 81N^4$ *members. Repeated phase patterns, if present, retain their induced weights.*

Proof. At any $k \leq 4$ distinct nodes the evaluation map from $\mathbb{F}_q^4$ onto $\mathbb{F}_q^k$ has rank k, since its first k columns form a Vandermonde matrix with nonzero determinant. Thus the corresponding polynomial values are independent uniform field elements. The trace $X + X^3 + \cdots + X^{3^{r-1}}$ is a nonzero $\mathbb{F}_3$-linear map $\mathbb{F}_q \to \mathbb{F}_3$: for $r > 1$ its nonzero polynomial has degree $3^{r-1} < q$, so cannot vanish everywhere, and for $r = 1$ it is the identity. It is consequently onto, with equal-sized fibers. The traced values are independent uniform ternary values at any at most four nodes. Every monomial in the second or fourth expansion involves at most four indices, with each index's net exponent between -2 and 2. Therefore the same cancellation argument applies. The leakage identity is quadratic and also remains exact. The nonlinear cutoff need not have the same distribution: its mean error bound only uses the already matched fourth moment. The finite-expectation selection proof and its probability estimate consequently apply without change. Minimality of r gives $q < 3N$, hence $q^4 < 81N^4$. □

One can search the smaller of the direct 3^N family and this field family. This proves a polynomial bound on the *number of candidate phase vectors*, not on total computation in an arbitrary von Neumann algebra. It leaves the number of orbit samples, the chosen averaging time, the evaluation of ϕ, and the implementation of polar and Fourier functional calculus unresolved. Limited-independent polynomial designs are classical; their exact compatibility with this nontracial certificate is the local application, not a claim to have invented the designs.

5 Explicit filters, bandwidth, and finite-time control

5.1 A compact trapezoidal multiplier

Fix real numbers $0 < a < b$, set $A = (a+b)/2$, $B = (b-a)/2$, and define

$$f_{a,b}(u) = \frac{1}{2B}(1_{[-A,A]} * 1_{[-B,B]})(u). \tag{5.1}$$

It equals one on $[-a, a]$, decreases linearly on each transition interval, vanishes outside $[-b, b]$, and lies in $[0, 1]$. It is continuous, not C_c^∞; Proposition 2.1 is why it is admissible.

Proposition 5.1 (Explicit Fourier cost and tail)**.** *With the Fourier convention of Section 2, an inverse kernel of (5.1) is*

$$g_{a,b}(t) = \frac{\sin(At)\sin(Bt)}{\pi B t^2}, \qquad g_{a,b}(0) = A/\pi. \tag{5.2}$$

It belongs to L^1, and

$$\|g_{a,b}\|_1 \le \sqrt{\frac{a+b}{b-a}}, \qquad \int_{|t|>R} |g_{a,b}(t)|\,dt \le \frac{4}{\pi(b-a)R} \quad (R>0). \tag{5.3}$$

For $\rho > 0$, $s \in \mathbb{R}$, the multiplier $f_{a,b}((u-s)/\rho)$ has kernel $g_{a,b;s,\rho}(t) = \rho e^{-ist} g_{a,b}(\rho t)$, with the same L^1 cost and tail at most $4/[\pi(b-a)\rho R]$.

Proof. Taking the negative-exponent Fourier transform of the two indicators and dividing by $2\pi(2B)$ gives (5.2). Near zero the removable limit is $AB/(\pi B) = A/\pi$; away from zero the kernel is $O(t^{-2})$, hence integrable. Fourier inversion for this integrable transform of a continuous integrable function proves the representation. Cauchy–Schwarz and the Plancherel identity $\int_{\mathbb{R}} (\sin(ct)/t)^2 dt = \pi c$ for $c > 0$ give

$$\int_{\mathbb{R}} |g_{a,b}(t)|dt \le \frac{\sqrt{\pi A}\sqrt{\pi B}}{\pi B} = \sqrt{A/B}.$$

The pointwise bound $|g_{a,b}(t)| \le 1/(\pi B t^2)$ gives tail $2/(\pi BR)$, equal to the stated expression. Change variables $v = \rho t$ for the rescaling assertions. The center phase has modulus one and does not affect these norms. □

For $a = 1, b = 2$ this gives the certified cost $\sqrt{3}$, without numerical integration of a chosen smooth bump. The bound is an upper bound, not a claim that the L^1 norm equals $\sqrt{3}$ or is minimal.

5.2 Arbitrarily small fixed enlargement

Corollary 5.2 (Width-flexible recovery consequence)**.** *In the source recovery construction, any prescribed fixed $\Lambda > 1$ can replace the factor four in (1.5). Specifically, put $\lambda = \sqrt{\Lambda}$ and use $f_{1,\lambda}$ for both localization stages. Once the averaging and finite-data step has supplied $a_T \ge \gamma$ and $C_4 \le C = 4 + e^{s+1}$, one may take*

$$K = \frac{\|T\|}{2}\sqrt{C/\gamma}, \qquad \eta = \frac{\sqrt{\gamma}}{2}, \qquad C_* = K\sqrt{\frac{\sqrt{\Lambda}+1}{\sqrt{\Lambda}-1}}. \tag{5.4}$$

The phase-selection step can use Proposition 4.4.

Proof. The first contraction filter fixes the initial radius δ_n and enlarges it to $d_n = \lambda\delta_n$. It maps $M\xi$ into $M\xi$ by Proposition 2.1, so the same density and normalization argument applies. Discard finitely many n until $d_n \le 1$; (3.4) and the same finite-data constant C follow from the audited averaging argument. Every phase sum has radius d_n. The second filter fixes that band and produces radius $\lambda d_n = \Lambda\delta_n$. In Theorem 4.2 take $L_f = \tau = 0$. Then

$$b \ge \sqrt{\gamma} - \frac{\|T\|\sqrt{C}}{4K} = \sqrt{\gamma}/2,$$

and Proposition 5.1 gives the operator bound in (5.4). The positive hypothesis ensures $T \ne 0$ and $K > 0$. All constants are independent of n. □

This corollary states precisely the consequence of the source's qualitative averaging inputs, not an independent verification of all global source claims. The reduced cutoff threshold is one fourth of the source's sufficient threshold at the same C, γ. More importantly, the recovery certificate now records the tradeoff between band width and algebra norm. As $\Lambda \downarrow 1$ the displayed norm bound diverges. Nothing here proves exact enlargement one or a uniform bound at that endpoint. The first filter's algebra norm does not enter C_* directly: its representatives may have large norm, which is absorbed only by an unquantified increase of the averaging time.

5.3 Finite time preserves a graph-norm certificate

Let $f(u) = f_{a,b}((u-s)/\rho)$, let g be its kernel, and truncate only time: $g_R = g1_{[-R,R]}$. For any $w \in M$ with $\|w\| \le K$, set $v = \mathcal{F}_g(w)$ and $v_R = \mathcal{F}_{g_R}(w)$, and put $\tau_R = \int_{|t|>R} |g(t)|dt$. Proposition 2.1 gives

$$\|v_R\| \le GK, \quad \|(v_R - v)\xi\| \le K\tau_R, \quad \left\|e^{D/2}(v_R - v)\xi\right\| \le K\tau_R. \tag{5.5}$$

The last bound uses $\left\|e^{D/2}w\xi\right\| = \|w^*\xi\| \le K$; it is not deduced from the first vector bound. The graph norm error is at most $\sqrt{2}K\tau_R$. In particular, setting $c = e^{s/2}$ and $d = b\rho$,

$$\|\phi v_R - e^s v_R \phi\|_{M_*} \le (1+c)K\{c(e^{d/2}-1) + (1+c)\tau_R\}. \tag{5.6}$$

Indeed, $v\xi$ has radius d and norm at most K, while $\left\|(e^{D/2} - c)(v_R - v)\xi\right\| \le (1+c)K\tau_R$; now apply Proposition 2.2.

If $\rho_n \to 0$, K stays bounded, and $\rho_n R_n \to \infty$, then (5.3) makes both terms in (5.6) tend to zero. Choosing, for instance, $R \ge 4/[\pi(b-a)\rho\tau]$ suffices for a prescribed tail tolerance $\tau > 0$. This is a sufficient time horizon, not a proved necessary or optimal one. Finite-time outputs lose exact compact spectral support; the useful replacement is the graph-norm and predual certificate.

5.4 What finite quadrature would additionally require

Time truncation leaves an integral, not a finite algorithm. On a partition $[-R, R] = \bigcup I_j$, choose $t_j \in I_j$, mesh diameter at most h, and weights $c_j = \int_{I_j} g(t)dt$. Define $\widehat{v} = \sum_j c_j \sigma_{t_j}(w)$. Then $\|\widehat{v}\| \le GK$ and

$$\begin{aligned} \|(\widehat{v} - v_R)\xi\| &\le G\,\omega_{w\xi}(h), \\ \left\|e^{D/2}(\widehat{v} - v_R)\xi\right\| &\le G\,\omega_{e^{D/2}w\xi}(h), \end{aligned} \tag{5.7}$$

where $\omega_v(h) = \sup_{|u|\le h} \|(U_u - I)v\|$. Integrate the pointwise difference $U_{t_j}v - U_t v$ to prove each bound; commutation with $e^{D/2}$ proves the second one. Each modulus tends to zero by strong continuity. A usable numerical rate needs additional information. For example, for any $L > 0$ the spectral theorem yields

$$\omega_v(h) \le hL\,\|v\| + 2\,\|E_D(\{|u| > L\})v\|\,. \tag{5.8}$$

One must control this tail for both $w\xi$ and $e^{D/2}w\xi$. The polar cutoff does not automatically preserve a bound for $Dw\xi$; therefore replacing these moduli by an unproved Lipschitz constant would be invalid. Errors in the scalar weights add their absolute sum times K to each vector bound. These statements explain exactly which extra certificates a genuinely numerical implementation would need.

6 Spectral rigidity and bicentralizer dependency assembly

6.1 From shrinking bands to an interval

For $a\xi = \xi a^\flat$ and fixed s, set $T_s = R(a^\flat) - L(b_s(a))$. If unit vectors on shrinking bands around s had nonvanishing averaged T_s-defect, recovery would produce uniformly bounded v_n with shrinking support and $\|T_s v_n \xi\| \geq \eta$. Equation (2.5) makes their predual defects tend to zero. The premise (1.6) instead forces

$$T_s v_n \xi = (v_n a - b_s(a) v_n)\xi \to 0,$$

a contradiction. This uses the hypothesis for the recovered bounded sequence, not for the unbounded initial representatives.

For a nonempty compact interval I, $c \in \mathbb{C}$, and $h \in E_D(I)H$, the source next establishes

$$\left\| t \mapsto (R(a^\flat) - cL(a))U_t h \right\|_{\mathrm{av}} \leq \|h\| \sup_{s \in I} \|(b_s(a) - ca)\xi\| . \tag{6.1}$$

To audit the passage, put $S = R(a^\flat) - cL(a)$, $C = P(S^*S)$, and let B be the supremum on the right. If a unit h violates the squared bound by $\varepsilon > 0$, partition I into Borel intervals of diameter at most $1/n$. Commutation of C with E_D removes cross terms, so one normalized spectral piece has quadratic value at least $B^2 + \varepsilon$. Compactness gives a subsequence supported in intervals shrinking to one $s \in I$. Decompose $S = T_s + L(b_s(a) - ca)$. The first term has averaged norm tending to zero by the point estimate; the second has squared averaged norm exactly $\phi((b_s(a) - ca)^*(b_s(a) - ca)) \leq B^2$ by (3.3). The triangle inequality contradicts the gap. For a singleton I use that same point. Thus no uniform-in-s rate has been assumed, and no operator-norm continuity of $b_s(a)$ was needed.

6.2 The symmetric pairing and its sign

State preservation gives the strongly continuous implementation $V_s(a\xi) = b_s(a)\xi$, $V_s = e^{isQ}$. It commutes with U_t, so D, Q strongly commute. Their joint spectral support $\mathcal{S} \subset \mathbb{R}^2$ has $E(\mathcal{S}) = 1$: the complement is a union of zero-projection open sets, reducible to a countable union using a countable base of $\mathbb{R}^2$. This point does not require separability of H.

For $(r, p), (q, \ell) \in \mathcal{S}$ choose nonzero $x\xi, a\xi$ in joint rectangles of half-width δ about those points. Such algebra vectors exist by density of $M\xi$ and successive weak Fourier filters along σ and b; a nonzero smaller rectangle ensures the filter product is nonzero. No uniform norm bound as $\delta \to 0$ is asserted. Let $c = e^{ir\ell}$, $d = e^{iqp}$, $H_0 = \|x\xi\| \, \|a\xi\| > 0$, and

$$F(t) = \sigma_t(x)a\xi, \qquad G(t) = a\sigma_t(x)\xi.$$

Equation (6.1) gives

$$\begin{aligned} \|F - cG\|_{\mathrm{av}} / H_0 &\leq \delta(|r| + |\ell|) + \delta^2 =: e_1, \\ \|G - dF\|_{\mathrm{av}} / H_0 &\leq \delta(|q| + |p|) + \delta^2 =: e_2. \end{aligned} \tag{6.2}$$

For the second line interchange a, x, reverse time, and use

$$U_t(\sigma_{-t}(a)x\xi - dx\sigma_{-t}(a)\xi) = G(t) - dF(t).$$

Crucially U_t is complex linear: d is not conjugated. Reflection invariance of m preserves the averaged seminorm. Moreover

$$\|F\|_{\mathrm{av}}^2 = \langle P(L(x^*x))a\xi, a\xi \rangle = H_0^2.$$

The identity $(1-cd)F = (F-cG)+c(G-dF)$ therefore yields

$$|1-e^{i(r\ell+qp)}| \le \delta(|r|+|\ell|+|q|+|p|)+2\delta^2. \tag{6.3}$$

Letting $\delta \to 0$ gives the source's symmetric relation $r\ell + qp \in 2\pi\mathbb{Z}$. The nonzero factor H_0 cancels before this limit; it need not be bounded below uniformly in δ.

If the two normalized estimates in (6.2) are known only up to additive errors $\varepsilon_1, \varepsilon_2$, the same proof adds their sum to (6.3). A fixed nonzero error does *not* give the exact lattice condition or the rigidity conclusion.

6.3 Spectral geometry and exclusion of modular eigenvalues

The bilinear form $\mathfrak{b}((r,p),(q,\ell)) = r\ell + qp$ is nondegenerate. If $\mathcal{S}$ contains independent v_1, v_2, then $z \mapsto (\mathfrak{b}(v_1,z), \mathfrak{b}(v_2,z))$ is invertible and sends $\mathcal{S}$ into $(2\pi\mathbb{Z})^2$; hence $\mathcal{S}$ is countable. Otherwise it lies in a line through zero. On a line $t(\alpha,\beta)$ with $\alpha\beta \neq 0$, self-pairing implies $\alpha\beta t^2 \in \pi\mathbb{Z}$, again a countable set. The other possibility is containment in a coordinate axis. This reasoning uses exact pairing, not merely numerically small phases.

For completeness, the source's modular eigenvalue exclusion extracts a bounded eigenoperator, rather than assuming a spectral projection preserves $M\xi$. If $E_D(\{r\})x\xi \neq 0$, the operators

$$y_k = \frac{1}{2k}\int_{-k}^{k} e^{-irt}\sigma_t(x)\,dt, \qquad \|y_k\| \le \|x\|,$$

have vectors converging by the unitary mean ergodic theorem to that nonzero projection. A weak operator cluster subnet gives $y \in M$ with that vector; separation implies $\sigma_t(y) = e^{irt}y$. Thus $y^*y, yy^* \in \mathbb{C}1$. Their positive scalar values coincide by equality of their operator norms, so y is a nonzero scalar multiple of a unitary. But (1.2) gives $\phi(yy^*) = e^r\phi(y^*y)$, whence $r = 0$. If D has pure point spectrum, this implies $D = 0$ and $M = M_\phi = \mathbb{C}1$.

Countable $\mathcal{S}$ makes its singleton spectral projections sum strongly to one, hence D has pure point spectrum. The vertical axis also gives $D = 0$. The horizontal axis gives $Q = 0$, hence $b_s(a)\xi = a\xi$ and $b_s(a) = a$. This completes the source's internal rigidity assembly conditional on the preceding hypotheses and inputs. It is not a finite matrix spectral test.

6.4 Where the global application imports theorems

The bicentralizer reduction invokes Houdayer–Isono's Proposition 3.4 and Theorem 3.5: a nonscalar bicentralizer B is an expected type III_1 factor and $\mathrm{BC}(B, \phi|_B) = B$ [4]. The restricted state is faithful normal, and separability passes to the expected subalgebra through the isometric predual embedding $\rho \mapsto \rho \circ E$. For a self-bicentralizing factor, every unitary in M_ϕ commutes with all of M, so scalarity of the factor center implies $M_\phi = \mathbb{C}1$.

The canonical bicentralizer flow theorem [5] gives

$$\|w_n\phi - \lambda\phi w_n\|_{M_*} \to 0 \implies w_n a - \beta_\lambda(a)w_n \to 0 \quad *\text{-strongly}.$$

The correct parameter is $b_s = \beta_{e^{-s}}$, because the predual expression in (1.6) is e^s times the negative of the one with $\lambda = e^{-s}$. Continuity and preservation of the restricted state are also required. Marrakchi's Theorem D supplies ergodicity for every nonidentity time, without a fullness assumption [6]. Rigidity makes this flow trivial; a trivial ergodic action has scalar algebra, contradicting the reduction.

These structural inputs are not reproved here. Similarly, the statement that a nontrivial algebra admitting a faithful normal scalar-centralizer state is type III_1 is an external structural conclusion, recorded in Marrakchi–Vaes Lemma 2.1 [2]; the eigenvalue exclusion alone is not substituted for its complete type-classification argument.

7 Specific Ram correspondences and their limits

All private passages cited here were read locally. The base directory for the following literal paths is

`C:/Users/mural/OneDrive/Documents/RamCalBooks/`

Their local use supplies representation conventions and certificate discipline, not an extra theorem about type III factors.

7.1 RamCS: the exact normalization bridge

The complete file `volumes/volume1-rev2/chapters/ch07-ramcs-rev2.tex`, "RamCS Transforms and Operator Spectra," was read with its seven input files. Its labels `eq:ramcs-definition`, `thm:ramcs-inversion`, `eq:ramcs-plancherel`, and `eq:ramcs-convolution` fix the positive-analysis convention

$$\mathcal{R}_c h(a) = \int_{\mathbb{R}} h(t) e^{a(t-c)} dt.$$

For the present kernel, $f(u) = \mathcal{R}_0 g(iu)$. The inverse formula is exactly $g(t) = (2\pi)^{-1} \int f(u) e^{-iut} du$ when its hypotheses hold. A nonzero time center c gives $\mathcal{R}_c g(iu) = e^{-iuc} f(u)$, which must be compensated in synthesis. The source's spectral center s instead modulates time by e^{-ist}; these are different parameters.

At the operator level the corresponding identity is $\mathcal{F}_g(y)\xi = f(D)y\xi$. It acts on the already defined modular spectral resolution. It does not identify H with scalar $L^2(\mathbb{R})$, assume an absolutely continuous modular spectrum, or diagonalize arbitrary $T \in \mathcal{B}(H)$. The same chapter's `eq:variable-plane-wave` explicitly distinguishes local symbols from eigenfunction relations. That warning prevents treating T_s as a scalar multiplier merely because the surrounding filters are functions of D.

The companion file `volumes/volume1-rev2/chapters/ch07-contours-C.tex`, subsection "Fourier and Laplace restrictions with all signs visible," explicitly identifies fixed-contour RamCS with weighted Fourier and bilateral Laplace transforms. Its subsection "What permits a contour shift" requires analyticity and horizontal-tail control. Accordingly, the factor $e^{D/2}$ in this report is handled by its spectral domain, not by an unproved imaginary-time shift of a weak integral.

7.2 Separate approximation budgets

The file `volumes/volume1-rev2/chapters/ch07-convergence-C.tex`, subsections "Truncated inversion and an explicit error norm" and "Space truncation and quadrature are separate errors," distinguishes normed tail errors from quadrature errors. This directly motivates the separate L_f, $\sqrt{C_4}/(4K)$, $K\tau$, and time quadrature terms in Sections 4–5. The actual inequalities here are proved in the state Hilbert norm, not imported from scalar pointwise error bounds.

The file `volumes/volume1-rev2/chapters/ch07-lab-rev2.tex`, "A polynomial multiplier with a certified remainder" and "A finite inverse filter certificate," stresses a whole-band estimate and a separate unretained-information remainder. The present certificate instead uses exact functional calculus plus measured leakage: a finite grid of modular energies by itself is not a uniform bound. The file `volumes/volume1-rev2/chapters/ch07-moments-C.tex`, "All ordinary moments need not give an exponential moment," provides a specific lognormal example. Its relevant lesson is that the fourth-moment bound used for polar cutoff must not be promoted to control of $e^{D/2}$ on arbitrary approximate vectors.

7.3 Domain transport and false finite conclusions

The full file `volumes/volume21/chapters/ch01a-state-domain.tex` was read, especially "The Object Behind a Formula," `v21c01:problem-record`, and "Transporting a Closed Realization," `v21c01:bounded-similarity`. Its closed-operator equivalence requires a bounded bijection with bounded inverse *and* the transported domain. Here $f(D)$ has compact spectral support and normally a kernel; it is a filter, not an invertible change of coordinates. Similarly $e^{D/2}$ is generally unbounded. Neither can be used as a bounded similarity on all of H. Volume 21 uses an inner product linear in the second variable; this report consistently retains the source's first-variable convention, rather than copying adjoint formulas without conjugating the convention.

The full file `volumes/volume21/chapters/ch01f-spectral-limits.tex` was read, including `v21c01:pollution-proposition`, `v21c01:strong-not-norm`, `v21c01:stronger-contract`, and the subsection "Domain-aware approximate inversion." Its example has a nested graph core and spurious compressed eigenvalue zero, with full residual $\sqrt{(N+1)^2 - 1/16}$. Thus even graph-core approximation and strong resolvent convergence do not justify global spectral exclusion. This applies directly to proposed tests of the joint support: a finite matrix model cannot establish the source's countability-or-axis dichotomy for the true joint spectral measure. Its domain-aware inverse criterion also requires complementary-space control; no finite-rank inverse is claimed here.

7.4 What is classical and what is contributed

The private Ram material provides a consistent representation of kernels, signs, domains, and error channels. Fourier inversion, the spectral theorem, Tomita–Takesaki theory, mean ergodic theorems, noncommutative randomization, and polynomial limited-independence are classical. Haagerup–Musat's fourth-moment and truncation work [3] and the source's cited Ioana–Vaes spectral-gap argument predate the present application. The contribution claimed here is the source-specific combination of finite phase selection, nontracial moment order, leakage tolerance, explicit width cost, and graph-aware time truncation, with proofs. No historical priority is asserted for those constituent inequalities or for their combination beyond this report's realized local work.

8 Current literature, comparisons, and open dependencies

Public primary literature was checked by public mathematical titles and arXiv identifiers on October 8, 2026 UTC; no private corpus text was used in public queries. Marrakchi–Vaes [2], final arXiv v3, supplies the scalar-centralizer/type-III_1 and weak-mixing context. Ando–Haagerup–Houdayer–Marrakchi [5] is the source of the canonical flow, and Marrakchi [6], Theorem D, explicitly states the ergodicity needed in the application. The more recent Houdayer–Marrakchi preprint has a v2 dated October 5, 2026, not just the v1 cited in the pinned source. Its bicentralizer conclusion is Theorem 9.11, and its resonance discussion is now Section 9.4 [7]. This is a current primary preprint claim, not external peer-review certification of this report or of the pinned manuscript. The v1 citations in the pinned source remain version-correct. In v2 the introductory absolute statement is Theorem C, with Theorem 9.11 its later formulation; its spectral-coordinate convention uses $e^{2\pi i p t}$. The present report retains e^{itu} and therefore retains $2\pi\mathbb{Z}$ in (6.3)'s limiting relation. Theorem numbers and Fourier normalization must be changed together when comparing versions.

Method	Advantage in this problem	Limitation or cost
Source smooth filters	Exact support and fixed L^1 cost under rescaling	Kernel constant unspecified; no finite averaging-time rate
Finite-phase certificate	Ordered moments checked exactly; finite deterministic witness family	Requires certified input moments, leakage, and filter evaluation
Field phase design	At most $q^4 < 81N^4$ candidate coefficient vectors	Does not bound orbit sample count or infinite-dimensional operation cost
Trapezoidal filters	Explicit support, cost, and time-tail bounds	Cost bound grows as transition width shrinks; only piecewise smooth
Time-truncated filters	Certified graph and predual errors	Exact compact spectral support is lost
Joint spectral assembly	Avoids uniform-in-energy pointwise rates	Uses exact limiting hypotheses and imported structural results

The main remaining dependencies are substantive. Effective recovery would need a bound on $\|\psi_k - \phi\| \, \|y_n\|^2$, a rate for $A_k(y_n^* y_n)\xi \to \xi$, quantitative continuity of the required orbit correlations, and computable bounds on both graph tails after cutoff. Scalar centralizer alone supplies the convergence but no such rates in the audited proof. Approximate numerical satisfaction of an intertwining law for finitely many operators is not its universal sequence quantifier. The source's bicentralizer reduction and canonical-flow theorems remain external dependencies, not consequences of a Ram transform identity.

Useful future work is therefore targeted: develop model-specific ergodic moduli; give validated finite representations with a complement estimate; optimize the transition-band filter for an actual permitted time horizon; and determine whether additional two-sided moment control reduces the graph-tail costs. Each project should preserve the separation between algebra norm, state Hilbert norm, and predual norm. No numerical speedup, new global classification theorem, or complete independent proof of bicentralizer triviality is claimed.

8.1 Why scalar centralizer cannot simply be dropped

The distinction between a finite-data certificate and uniform production of that data is necessary, not just cautious terminology. Consider the bounded direct product $M = \prod_{m \geq 1} M_2(\mathbb{C})$ with faithful normal state

$$\phi((x_m)) = \sum_{m \geq 1} 2^{-m} \operatorname{Tr}(\rho_m x_m), \qquad \rho_m = \frac{\operatorname{diag}(e^{r_m}, 1)}{1 + e^{r_m}}, \quad r_m = 1 + \frac{1}{m}. \tag{8.1}$$

The state is normalized since $\sum_{m \geq 1} 2^{-m} = 1$, and faithful since every density is strictly positive. The modular action is blockwise conjugation by ρ_m^{it}. Its centralizer contains all diagonal blocks. The vector $e_{12}^{(m)}\xi$ has energy r_m and squared norm $2^{-m}/(1 + e^{r_m})$. Normalize it to a unit h_m, take $T = I$, $s = 1$, and $\delta_n = 1/n$. Then h_n has the required input band and $m_t \, \|U_t h_n\|^2 = 1$ for any normalized mean.

For $n > 4$, however, an algebra vector $v_n\xi$ supported in $[1 - 4/n, 1 + 4/n]$ can contain only the e_{12} entry in blocks $m \geq \lceil n/4 \rceil$: the zero and negative energies are excluded, and the positive energy $1 + 1/m$ belongs to the band exactly under that condition. The strictly positive densities make all other entries of v_n vanish. Therefore $\|v_n\| \leq C$ implies

$$\|v_n\xi\|^2 \leq C^2 \sum_{m \geq \lceil n/4 \rceil} 2^{-m} = C^2 2^{1 - \lceil n/4 \rceil} \longrightarrow 0. \tag{8.2}$$

No positive uniform output mass is possible, even along a subsequence. Thus the source recovery conclusion fails without a replacement for its scalar-centralizer hypothesis. This example concerns the stated class of von Neumann algebras; it makes no separate claim about every narrower class of factors.

8.2 Nonzero versus nonscalar witnesses

Recovery ensures $\|v_n\xi\| \geq \eta/\|T\| > 0$. Normalizing these vectors preserves a uniform algebra bound, at most $C_* \|T\|/\eta$, but does not make v_n unitary or identify it with the original h_n. For $s \neq 0$ the shrinking output bands eventually omit zero; since $D\xi = 0$, their vectors are orthogonal to ξ, so $\phi(v_n) = 0$ and the witnesses are nonscalar.

At $s = 0$ this does not follow: $M = \mathbb{C}$, $T = I$, and $h_n = \xi$ satisfy the positive hypothesis with only scalar outputs. A precise repair is to use $T\xi = 0$, for example $T = I - P_{\mathbb{C}\xi}$. Centering then gives $u_n = v_n - \phi(v_n)1$ with the same zero-centered band, $Tu_n\xi = Tv_n\xi$, $\|u_n\| \leq 2C_*$, and $\|u_n\xi\| \geq \eta/\|T\|$. In the actual source point argument $T_0\xi = a\xi - b_0(a)\xi = 0$, so this safeguard is compatible with its application, although the contradiction only needs positive T_0-mass.

Finally, shrinking bands with nonvanishing vector norms do not produce a nonzero weak limit. Any weak operator cluster point of a uniformly bounded sequence on bands shrinking to s has its vector in $E_D(\{s\})H$, as is seen by testing against vectors supported outside successive neighborhoods of s. For scalar centralizer and $s \neq 0$, the eigenvalue exclusion forces the cluster point to vanish. At $s = 0$, the Hilbert mean ergodic limit and density give $E_D(\{0\})H = \mathbb{C}\xi$; centered cluster points vanish there as well. This is a concrete obstruction to replacing the source's bounded sequence by a nonzero limiting eigenoperator.

9 Verification and contribution boundary

The analytical proofs in Sections 2, 4, and 5 establish local statements with explicit hypotheses. The audit in Sections 3 and 6 records source dependencies and checks the quantifier order, noncommutative factor order, reflection sign, and domain usage. It does not replace an independent review of the imported literature.

Finite tests use a faithful nontracial state on $M_2(\mathbb{C})$, its standard Hilbert–Schmidt representation, exact enumeration of a small ternary family, and explicit modular eigenvalues. They are algebraic and floating-point consistency checks, not tests of scalar-centralizer recovery: no nontrivial finite-dimensional factor with a faithful normal state has scalar centralizer. Exact symbolic identities and analytic bounds carry the general proofs. The accompanying local verification record specifies executed tests, tolerances, observed residuals, source hashes, and full source-read scope. Nothing is inferred from unexecuted experiments or from a visual fit.

The nontracial matrix diagnostic used $\rho = \mathrm{diag}(0.3, 0.7)$, $\xi = \rho^{1/2}$, $D_{ij} = \log\rho_i - \log\rho_j$, five explicit matrices, probability weights $(0.1, 0.15, 0.2, 0.25, 0.3)$, and a fixed Hilbert–Schmidt-space operator normalized to norm one. At $K = 3$, spectral center 0.2, and scale 0.65, all $3^5 = 243$ phases were enumerated. The recorded values were

$$a_T = 0.3810422031, \quad C_4 = 1.91101495, \quad L_f = 0.4100694125, \qquad b = 0.0920171937.$$

The observed output RMS was 0.5167922251, and the largest output norm was 0.7625557527. These values verify this manufactured example's certificate; they do not estimate an optimal constant. The fourth-moment matrix identity had residual 1.87×10^{-15} against tolerance 10^{-11}; second-moment and leakage residuals were below 1.4×10^{-16}. The tested cutoff and predual inequalities had no positive violation above 10^{-12}.

Exact integer enumeration over $\mathbb{F}_9 = \mathbb{F}_3[\theta]/(\theta^2 + 1)$ checked 6561 coefficient seeds. The five-node check verified every nonempty marginal of size at most four, 30 subsets in total. An eight-node check verified all 70 four-node marginals and all 4096 indexed fourth-moment patterns. Each nontrivial phase exponent had exactly equal ternary counts. The scalar cutoff check also verified the exact factorization

$$x^4 - 16K^2(x - K)^2 = (x - 2K)^2(x^2 + 4Kx - 4K^2),$$

whose final factor is positive for $x \geq K > 0$, with equality of the cutoff bound at $x = 2K$. These exact finite checks complement, rather than replace, the all-size proofs.

The internal contribution assessment is C3/5. Its basis is the finite nontracial recovery theorem, the polynomial-size phase realization, and the combined bandwidth and graph-norm error contract. These give a meaningful local extension of the source's existence step. The number 1/16 alone would not support that assessment. The source's headline global result, an effective rate for ergodic averaging, historical novelty, and a measured end-to-end computational improvement are not realized contributions of this report.

References

[1] OpenAI, *Bounded recovery for modular spectral averages*, September 23, 2026, catalogue ID M580, pinned commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Local eight-file TeX source described in Section 1.

[2] A. Marrakchi and S. Vaes, *Ergodic states on type* III_1 *factors and ergodic actions*, J. Reine Angew. Math. 809 (2024), 247–260. Theorem A and Lemma 2.1. https://arxiv.org/abs/2305.14217v3.

[3] U. Haagerup and M. Musat, *On the best constants in noncommutative Khintchine-type inequalities*, J. Funct. Anal. 250 (2007), 588–624. Equation (2.10), Lemma 2.6, and Remark 2.8. https://arxiv.org/abs/math/0611160v3.

[4] C. Houdayer and Y. Isono, *Unique prime factorization and bicentralizer problem for a class of type III factors*, Adv. Math. 305 (2017), 402–455. Proposition 3.4 and Theorem 3.5. https://arxiv.org/abs/1503.01388v3.

[5] H. Ando, U. Haagerup, C. Houdayer, and A. Marrakchi, *Structure of bicentralizer algebras and inclusions of type III factors*, Math. Ann. 376 (2020), 1145–1194. Theorem A. https://arxiv.org/abs/1804.05706v2.

[6] A. Marrakchi, *Full factors, bicentralizer flow and approximately inner automorphisms*, Invent. Math. 222 (2020), 375–398. Theorem D. https://arxiv.org/abs/1811.10253v2.

[7] C. Houdayer and A. Marrakchi, *The classification of flows on* II_1 *factors and Connes' bicentralizer problem*, arXiv:2609.11462v2, October 5, 2026, 150 pages. Section 9.4 and Theorem 9.11. https://arxiv.org/abs/2609.11462v2.

[8] M. Subbarao, private Ram Calculus corpus, Volume 1 revised 2, Chapter 7 and companion files; Volume 21, Chapter 1 files `ch01a-state-domain.tex` and `ch01f-spectral-limits.tex`. Exact local paths, headings, and labels are given in Section 7. These are private representation sources, not public literature claims.